

ML Week 6

Number of participants: 31



1. I watch the course videos before the lecture:

14 correct answers
out of 20 respondents

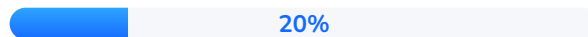


Almost always



14 votes

Sometimes



4 votes

Almost never



2 votes



2. The content of the videos is

17 correct answers
out of 25 respondents



The right level for me to understand it



17 votes

Too easy



0 votes



3. The explanations during the lectures are

18 correct answers
out of 24 respondents



Helpful to better understand the content of the videos



18 votes

Too difficult to follow



4 votes

I can follow but not helpful



2 votes



4. On SpeakUp:

1 correct answer
out of 22 respondents



I regularly ask my questions if I have any



1 vote

I have not understood everything but I don't know what to ask



11 votes

I hope someone else asks my question



5 votes

I don't have time to use it



3 votes

SpeakUp ???



2 votes



5. Image Recognition

0 correct answer
out of 23 respondents



There is additional information in images that we have not used in the current methods.



39%

9 votes

The more pixels an image has, the more difficult is the classification problem.



30%

7 votes



In the handwritten digits data set, the training samples are almost linearly separable.



22%

5 votes



Feature engineering can help to improve the prediction error.



74%

17 votes



6. Classification and Regression Trees

3 correct answers
out of 24 respondents



Trees can have
zero training error.



16 votes

Larger trees have
smaller prediction
variance.



7 votes

Trees are more
flexible than linear
models.



18 votes

Trees are
generative models,
that is, they model
the joint
distribution
 (X, Y) .



6 votes



7.

Which statement(s) hold(s) in general for the cost-complexity pruning parameter α for an initially unpruned classification tree?

4 correct answers
out of 18 respondents



$\alpha = 0$ always
yields zero training
error.



11 votes

Increasing α
decreases the bias.



7 votes

α is chosen with
the Gini index or
the entropy.



5 votes



Typically, the test
error first
decreases and
then increases, as
 α increases.



13 votes



8. Decision trees.

5 correct answers
out of 17 respondents



They are interpretable models.



12 votes

Their computational complexity does not depend on the dimension of the predictor space.



4 votes



They are simple to fit in multiclass classification problems.



10 votes



Can always achieve zero misclassification error on the training data set.



15 votes

